

Customer Churn Prediction

Exploratory Data Analysis (EDA) Report

Telco Customer Churn Dataset

Machine Learning Project

1. Introduction

Customer churn refers to customers discontinuing their services with a company. Understanding customer churn helps organizations identify customers who may leave and develop effective customer retention strategies.

This Exploratory Data Analysis was performed on the Telco Customer Churn dataset to understand customer characteristics and identify important patterns associated with churn.

2. Dataset Overview

Parameter	Value
Total Customers	7043
Total Features	21
Churned Customers	1869
Non-Churned Customers	5174
Churn Percentage	26.54%
Non-Churn Percentage	73.46%
Duplicate Rows	0

The dataset contains 7,043 customer records and 21 attributes. Approximately 26.54% of customers have churned, while the remaining customers have continued their services.

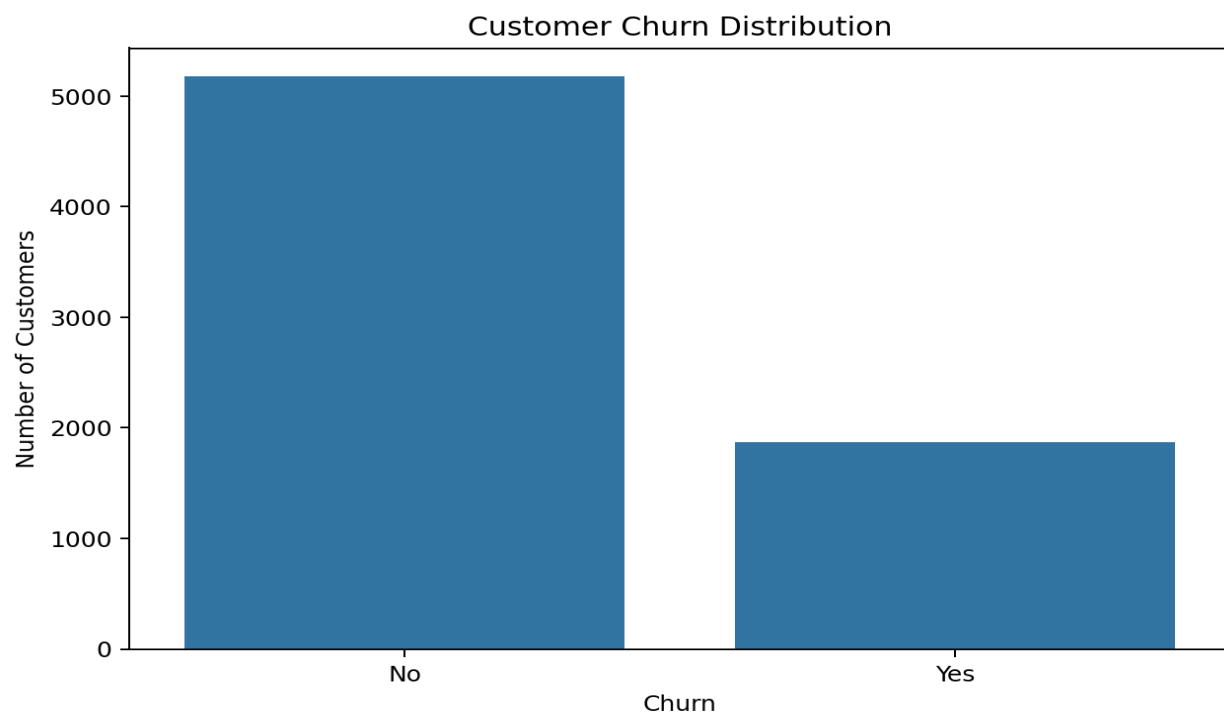
3. Data Quality Analysis

The dataset was examined for duplicate records and missing values. The number of duplicate rows identified was 0. A total of 11 missing values were identified after data inspection.

The TotalCharges column was initially stored as an object/string data type and was converted into numerical format for analysis. Any invalid or blank values encountered during conversion were treated as missing values and should be handled during the data-preprocessing stage.

4. Churn Distribution

Out of 7,043 customers, 1,869 customers (26.54%) churned, while 5,174 customers (73.46%) did not churn.

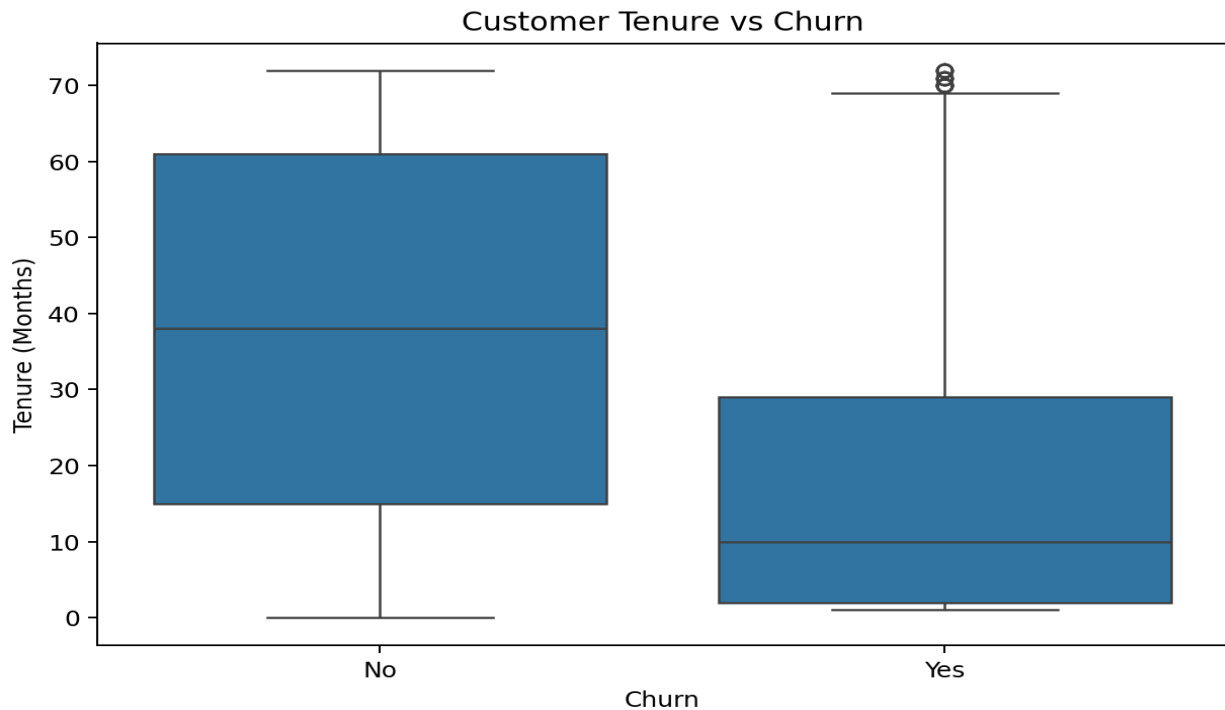


The target variable is therefore somewhat imbalanced, with non-churned customers representing the majority class.

5. Numerical Feature Analysis

5.1 Tenure vs Churn

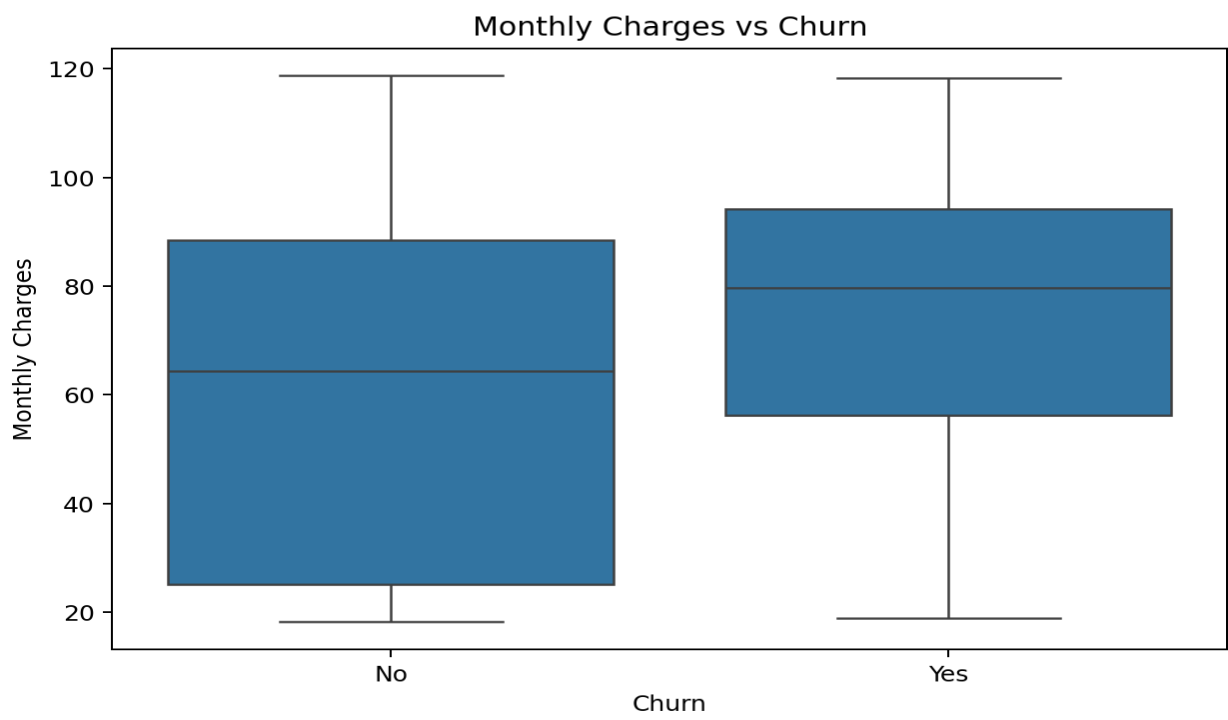
The average customer tenure is approximately 32.37 months. The boxplot indicates that churned customers generally have shorter tenure compared with customers who remain with the company.



Tenure has a correlation of -0.352 with churn, making it the strongest numerical feature associated with the target variable.

5.2 Monthly Charges vs Churn

The average monthly charge is approximately 64.76. The analysis indicates that customers with higher monthly charges tend to show higher churn tendencies.

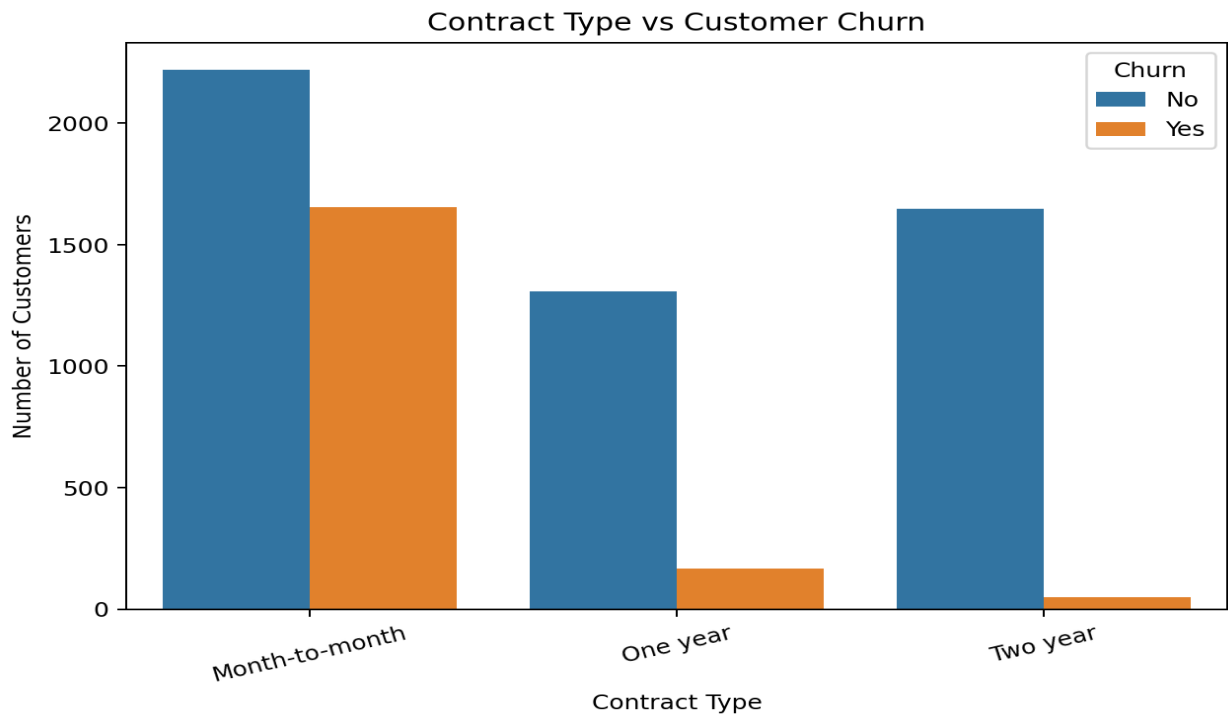


MonthlyCharges has a correlation of 0.193 with churn.

6. Categorical Feature Analysis

6.1 Contract Type vs Churn

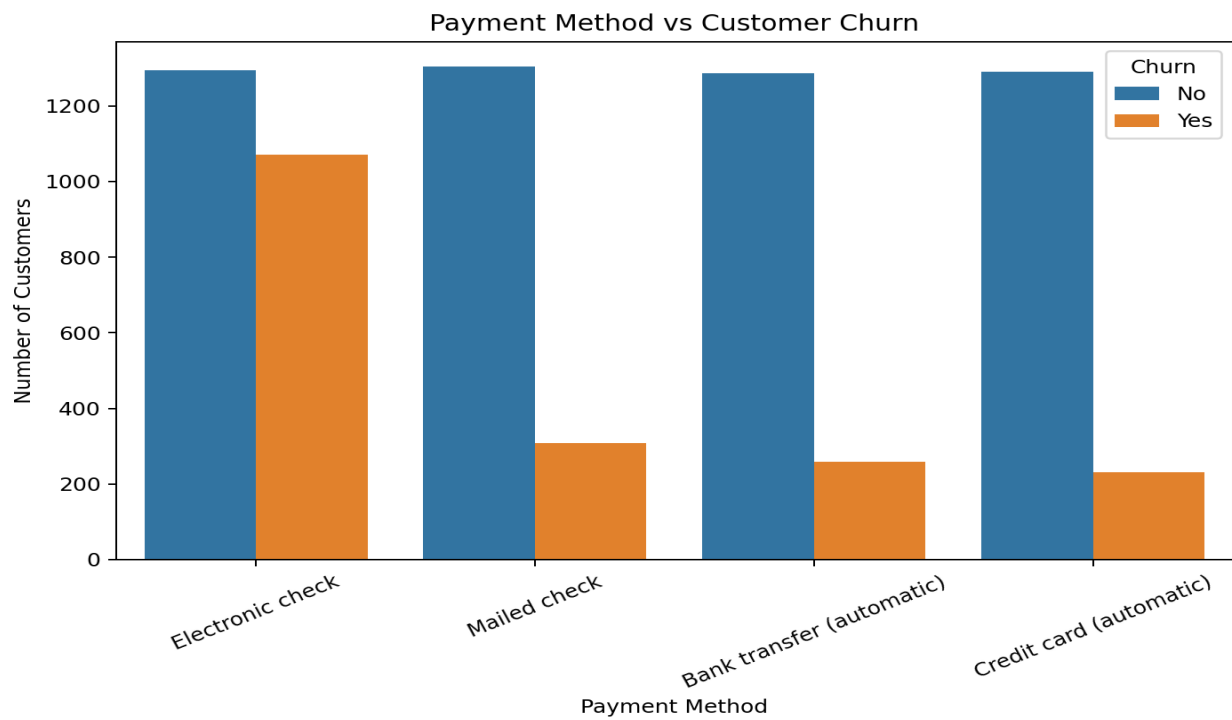
Contract type shows an important relationship with customer churn. Month-to-month customers generally show considerably higher churn than customers with one-year or two-year contracts.



Contract Type	No Churn (%)	Churn (%)
Month-to-month	57.29	42.71
One year	88.73	11.27
Two year	97.17	2.83

6.2 Payment Method vs Churn

Payment method also shows noticeable differences in churn rates. Electronic check has the highest churn rate among the payment methods analyzed.



Payment Method	No Churn (%)	Churn (%)
Bank transfer (automatic)	83.29	16.71
Credit card (automatic)	84.76	15.24
Electronic check	54.71	45.29
Mailed check	80.89	19.11

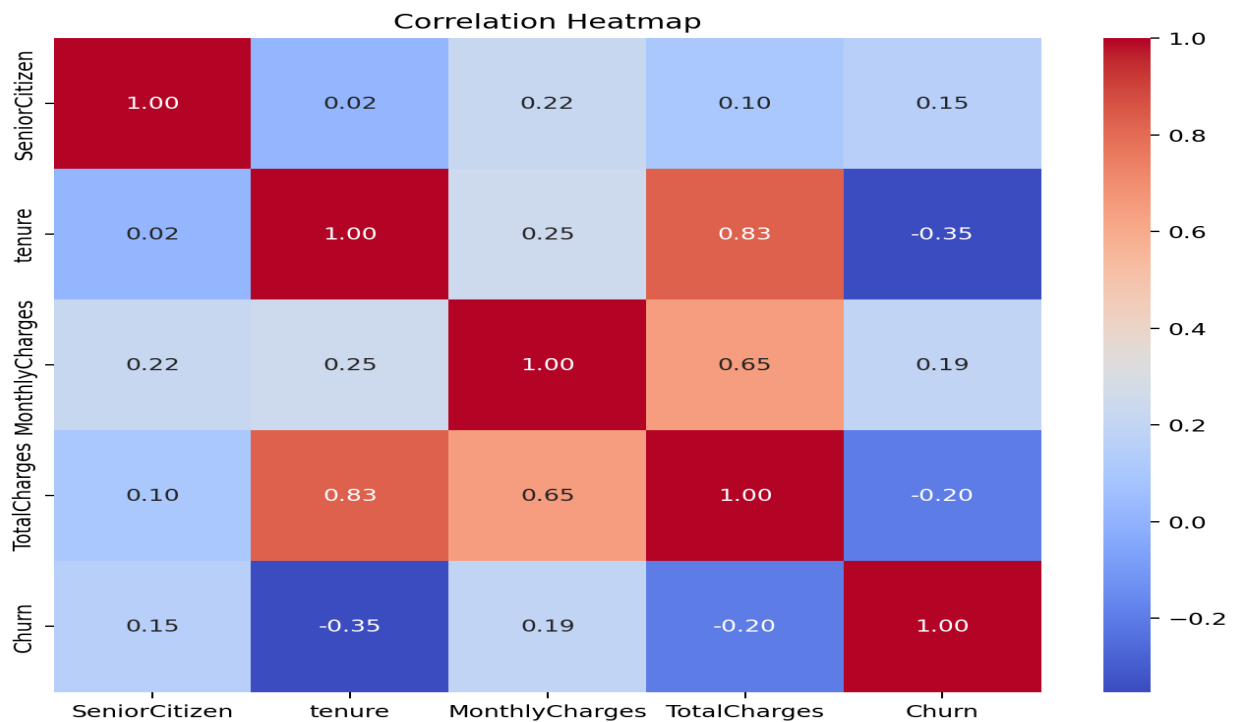
6.3 Partner Status vs Churn

Customers without a partner have a churn rate of 32.96%, compared with 19.66% for customers with a partner. This indicates that partner status shows a noticeable association with customer churn.

7. Correlation Analysis

Correlation analysis was performed to identify relationships between numerical features and the churn target variable.

Feature	Correlation with Churn
tenure	-0.352
TotalCharges	-0.199
MonthlyCharges	0.193
SeniorCitizen	0.151



8. Key EDA Findings

- Approximately 26.54% of customers have churned.
- Tenure has the strongest numerical relationship with churn (-0.352).
- Customers with shorter tenure tend to have higher churn.
- MonthlyCharges has a positive correlation with churn (0.193).
- Month-to-month contract customers show considerably higher churn than customers with longer contracts.
- Electronic check customers have a churn rate of 45.29%.
- Customers without partners have a churn rate of 32.96%, compared with 19.66% for customers with partners.
- Gender shows only a small difference in churn and therefore does not appear to be a major churn indicator in this analysis.

9. Conclusion

The exploratory analysis identifies several important patterns associated with customer churn. Tenure, contract type, monthly charges, payment method, and partner status show meaningful relationships with churn.

The findings from this EDA provide a foundation for the next stage of the project. The cleaned and appropriately encoded data can subsequently be used to train machine learning models such as Random Forest and XGBoost for customer churn prediction.